DUY LE HAN, Ph.D. (TS. Hán Lê Duy)

Lecturer & Researcher | Vice Director | General Secretary

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Department of Welding Engineering and Metals Technology
Joining and Welding Research Institute (JWRI HUST-UOsaka)
School of Mechanical Engineering (SME), Hanoi University of Science and Technology (HUST)
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- **Google Scholar:** Duy Le Han
- **Date of Birth:** May 17, 1987 | **Nationality:** Vietnamese
- **Languages:** Vietnamese (Native), English (Fluent), Japanese (Intermediate)

1. EDUCATION

- **2018 – 2022: Ph.D. in Materials and Manufacturing Science**
  - **Institution:** Joining and Welding Research Institute (JWRI), Osaka University, Japan
  - **Research Focus:** Low-temperature lead-free solder alloys, interfacial microstructure evolution, and solid-state nanoporous copper bonding for microelectronics and power device packaging.
- **2013 – 2015: Master of Engineering (M.Eng.) in Mechanical Engineering**
  - **Institution:** School of Mechanical Engineering, Hanoi University of Science and Technology (HUST), Vietnam
  - **Thesis:** "Study about welding engineering and technology of heat-resistant steel"
- **2005 – 2010: Bachelor of Engineering (B.Eng.) in Welding Engineering**
  - **Institution:** School of Mechanical Engineering, Hanoi University of Science and Technology (HUST), Vietnam

2. PROFESSIONAL CERTIFICATIONS & SPECIALIZED TRAINING

- **2024 – 2025: Certified Expert in Energy Management Systems (ISO 50001)** — Jointly organized by the Ministry of Industry and Trade (MOIT) and the United Nations Industrial Development Organization (UNIDO).
- **2024 – 2025: Certificate in Leadership and Departmental Management** (Decision No. 94/QĐ-CBQL).
- **April 2017: Certified Assessor and Examiner** — Asian Welding Federation (AWF).

- **2012 – 2015: Certified Welding Inspector (AWS CWI)** — American Welding Society (Cert. No. 15080521).
- **2012 – 2015: International Welding Inspector – Standard (IIW IWI-S)** — International Institute of Welding (Diploma No. VN-SL-Hanoi-1178-111226-008-IIW).
- **February 2015: Radiographic Testing (RT) Film Interpretation Certificate**.
- **February 2014: Canadian Welding Engineer Training Program**.

3. ACADEMIC & PROFESSIONAL APPOINTMENTS

- **2024 – 2029: General Secretary**
    - *Vietnam Welding Society (VWS), Hanoi, Vietnam*
    - Coordinating industry-academia linkages, fostering international collaborations, and developing the national professional welding community.
- **December 2023 – Present: Vice Director**
    - *Research Center for Welding & Joining Technology, School of Mechanical Engineering, HUST*
    - Leading scientific research initiatives, industrial technology transfer, and consulting on advanced welding and joining engineering solutions.
- **March 2011 – Present: Lecturer & Researcher**
    - *Department of Welding Engineering & Metals Technology, School of Mechanical Engineering, HUST*
    - Teaching undergraduate and graduate courses, supervising B.Eng./M.Sc./Ph.D. theses, academic advising, and conducting international welding certification training (IIW/AWS).
- **2018: Technical & Training Expert**
    - *Pilot Training Program Support Project (TV05B) — Directorate of Vocational Education and Training (MOLISA)*
    - Implemented welding technician training programs in accordance with French national standards.
- **2012 – 2016: Head of Training Department & Head of Business Division**
    - *CTWEL Consulting and Training JSC, Hanoi, Vietnam*
    - Designed and managed welding personnel training programs according to AWS standards; provided technical consulting on welding engineering and ISO 3834 quality management systems.
- **2009 – 2010: Welding Technician / Intern**
    - *Tan The Ky Co., Ltd., Hanoi, Vietnam*
    - Installed and commissioned industrial welding equipment, power sources, and robotic welding systems.

4. RESEARCH INTERESTS

- **Advanced Soldering & Microelectronics Packaging:** Low-temperature lead-free solder alloys (In–Sn–Cu, In–Sn–Ag), nanoparticle/whisker-reinforced composite solders (SAC-ZrO2, SAC-ZnO), solid-state nanoporous Cu bonding, and electromigration reliability.
- **Advanced Welding & Joining Technologies:** Friction Stir Welding (FSW), novel GTAW processes for ultra-thin metal sheets, Plasma Arc Welding (PAW), and welding metallurgy of heat-resistant steels (SA335 P91) and quenched & tempered steels.
- **In-situ Characterization & Nondestructive Evaluation:** Real-time "In-situ" X-ray imaging for materials and manufacturing processes, microstructural characterization, radiographic testing (RT), and welding quality assurance (ISO 3834).

5. TEACHING & PROFESSIONAL TRAINING

University Courses (HUST)

- Welding Processes and Welding Engineering
- Brazing and Soldering Technology & Equipment
- Welding Quality Assurance
- Heat Treatment of Welded Joints

International Certification Training

- Authorized Instructor for **International Institute of Welding (IIW)** and **American Welding Society (AWS)** personnel qualification and certification programs across multiple levels.

6. FUNDED RESEARCH PROJECTS

Role	Project Code & Details	Topic
Key Member	Ministry-Level Project(Code: B2023-BKA-12)	Research on establishing technological procedures and fabricating an automated welding system for butt-welding of dissimilar thermoplastic pressure pipes.
Principal Investigator	Young Ph.D. Grant(Code: T2022-TT-006)	Study on the effect of isothermal aging temperature on the mechanical strength of low-temperature lead-free solders.
Principal Investigator	University-Level Project(Code: T2015-211)	Study on welding technology of SA335 P91 heat-resistant alloy steel used in thermal power plants.

Role	Project Code & Details	Topic
Principal Investigator	University-Level Project(Code: T2014-34)	Study on welding technology of quenched and tempered carbon steel, applied to welding Sandvik 1300C conveyor steel belts.
Principal Investigator	University-Level Project(Code: T2012-21)	Research, design, and fabrication of a pneumatic pressure generating device for spot welding robots.

7. PUBLICATIONS

A. Peer-Reviewed International Journal Articles (ISI / Scopus)

1. Bui Van Hanh, Trinh Quang Ngoc, **Han Le Duy**, Phan Huy Le, Effect of welding parameters on void defect formation in friction stir welding of 6061 aluminum alloys, International Journal of Advanced Manufacturing Technology (2026), Q1, IF=3.563, DOI: https://link.springer.com/10.1007/s00170-026-17807-1
2. Bui Van Hanh, Trinh Quang Ngoc, Le Dang Khoi, Tashiro Shinichi, **Han Le Duy**, Le Phan Huy, Murphy Anthony B., Yamanaka Kenta, Tanaka Manabu, Xiao Lei, Effect of external magnetic field on arc characteristics and weld bead formation in metal-cored arc welding, International Journal of Advanced Manufacturing Technology (2025), Q1, IF=3.563, DOI: https://link.springer.com/10.1007/s00170-025-16581-w
3. Van Anh Nguyen, Dilen Damian, Quang Ngoc Trinh, **Han Le Duy**, Tien-Dat Hoang, Dung Hoang Tien, Developing novel arc plasma spot welding process, CHINA WELDING (2025), Q2, IF=1.054, DOI: 10.12073/j.cw.20250412025
4. Van Anh Nguyen, **Duy Han Le**, Dilen Damian, The Bach Tran, Quang Hung Le, Nhu Tung Nguyen, “In-situ” x-ray imaging technology for material and manufacturing science: A review, Journal of Manufacturing Processes (2025), Q1, IF=6.2, DOI: https://doi.org/10.1016/j.jmapro.2025.03.049
5. **Duy Le Han**, Bui Van Hanh, Nguyen Van Anh, Trinh Quang Ngoc, Hiroshi Nishikawa, Properties of In-Sn-xCu Alloys and Their Joints on a Cu Substrate Before and After Isothermal Aging, Materials Science and Engineering: A (2025), Q1, IF=6.1. DOI: https://doi.org/10.1016/j.msea.2025.147796
6. Zhi Jin, Fupeng Huo, **Duy Le Han**, Xunda Liu, Hiroaki Tatsumi, YC Chan, Hiroshi Nishikawa, The electromigration study of thin-film structured Sn3. 5Ag and Ag using continuous observation and reliability enhancement approach, Thin Solid Films (2023), Q2, IF=2.1. DOI: https://doi.org/10.1016/j.tsf.2023.139827
7. Ngo Huu Manh, Van Thao Le, **Duy Le Han**, V-A Nguyen, Successful joining of ultra-thin AA3003 aluminum alloy sheets by the novel GTAW process, Vacuum (2022), Q1, IF=4.0. DOI: https://doi.org/10.1016/j.vacuum.2022.111558
8. Fupeng Huo, Zhi Jin, **Duy Le Han**, Jiahui Li, Keke Zhang, Hiroshi Nishikawa, Novel interface regulation of Sn1. 0Ag0. 5Cu composite solders reinforced with modified ZrO2: Microstructure

and mechanical properties, Journal of Materials Science & Technology (2022), Q1, IF=10.9. DOI: https://doi.org/10.1016/j.jmst.2022.01.040

9. Ngo Huu Manh, Van Anh Nguyen, **Han Le Duy**, Murata Akihisa, Van Thao Le, Trinh Quang Ngoc, Bharat Gandham, Development of a novel GTAW process for joining ultra-thin metal sheets, Journal of Manufacturing Processes (2022), Q1, IF=6.2. DOI: https://doi.org/10.1016/j.jmapro.2022.06.043
10. Byungho Park, **Duy Le Han,** Mikiko Saito, Jun Mizuno, Hiroshi Nishikawa, Effect of various parameters on the shear strength of solid-state nanoporous Cu bonding in Cu–Cu disks for power device packaging, Journal of Electronic Materials (2022), Q2, IF=2.1. DOI: https://doi.org/10.1007/s11664-022-09634-3
11. **Duy Le Han,** Yu-an Shen, Siliang He, Hiroshi Nishikawa, Effect of Cu addition on the microstructure and mechanical properties of In–Sn-based low-temperature alloy, Material Science & Engineering A, 804 (2021) 140785, Q1, IF=6.4. DOI: https://doi.org/10.1016/j.msea.2021.140785
12. **Duy Le Han**, Yu-An Shen, Fupeng Huo, Hiroshi Nishikawa, Microstructure Evolution and Shear Strength of Tin-Indium-xCu/Cu Joints, Metals (2021), Q2, IF=2.9. DOI: https://doi.org/10.3390/met12010033
13. Fupeng Huo, Zhi Jin, **Duy Le Han**, Keke Zhang, Hiroshi Nishikawa, Interface design and the strengthening-ductility behavior of tetra-needle-like ZnO whisker reinforced Sn1. 0Ag0. 5Cu composite solders prepared with ultrasonic agitation, Materials & Design, 210 (2021), Q1, IF=8.4. DOI: https://doi.org/10.1016/j.matdes.2021.110038
14. Byungho Park, **Duy Le Han**, Mikiko Saito, Jun Mizuno, Hiroshi Nishikawa, Fabrication and characterization of nanoporous copper through chemical dealloying of cold-rolled and annealed Mn–Cu alloy, Journal of Porous Materials (2021), Q2, IF=2.6. DOI: https://doi.org/10.1007/s10934-021-01128-7
15. Huu Loc Nguyen, Anh Van Nguyen, **Han Le Duy**, Thanh-Hai Nguyen, Shinichi Tashiro, Manabu Tanaka, Relationship among Welding Defects with Convection and Material Flow Dynamic Considering Principal Forces in Plasma Arc Welding, Metals, 2021, Q2, IF=2.9. DOI: https://doi.org/10.3390/met11091444
16. **Duy Le Han,** Yu-an Shen, Sanghun Jin, Hiroshi Nishikawa, Microstructure and mechanical properties of the In–48Sn–xAg low-temperature alloy, Journal of materials science, 0022-2461 (2020), Q1, IF=4.5. DOI: https://doi.org/10.1007/s10853-020-04691-7

B. International & National Conference Proceedings

1. **Duy Le Han**, Hiroaki Tatsumi, Fupeng Huo, Hiroshi Nishikawa (2022). Effect of isothermal aging on properties of In-48Sn and In-Sn-8Cu alloys. *2022 IEEE 72nd Electronic Components and Technology Conference (ECTC)*. DOI: 10.1109/ECTC51906.2022.00339
2. **Duy Le Han**, Byungho Park, Hiroshi Nishikawa (2021). Effect of 4.0 mass % Cu Addition on Microstructure and Mechanical Properties of In-48Sn Alloy. *International Conference on Electronics Packaging (ICEP 2021)*, Japan.
3. Byungho Park, **Duy Le Han**, Mikiko Saito, Jun Mizuno, Hiroshi Nishikawa (2021). The Effect of Solid-state Nanoporous Cu Bonding for Power Device. *International Conference on Electronics Packaging (ICEP 2021)*, Japan.

4. Chau Tran Van, **Duy Han Le**, Tung Do Thanh (2017). Effect of rotating speed on historical temperature and tensile strength of Friction Stir Welding AA 5052 Joint. *The 3rd International Workshop on Welding and Joining (IWWJ 2017)*.

C. National Journal Articles

1. Tran Van Chau, **Han Le Duy**, Do Thanh Tung (2018). Study on the effect of rotational speed on thermal regime and tensile strength of friction stir welded AA5052 aluminum alloy joints (Nghiên cứu ảnh hưởng của tốc độ quay đến chế độ nhiệt và độ bền kéo của liên kết hàn ma sát ngoáy hợp kim nhôm 5052). *Vietnam Mechanical Engineering Journal (Tạp chí Cơ khí Việt Nam)*, ISSN 0866-7056.
2. **Han Le Duy** (2017). Study on welding technology of P91 alloy steel by Shielded Metal Arc Welding (SMAW) and Gas Tungsten Arc Welding (GTAW) (Nghiên cứu công nghệ hàn thép hợp kim P91 bằng phương pháp hàn hồ quang tay và hàn hồ quang bằng điện cực không nóng chảy trong môi trường khí bảo vệ). *Journal of Science and Technology of Technical Universities (Tạp chí Khoa học & Công nghệ các Trường Đại học Kỹ thuật)*, No. 117, pp. 049–053, ISSN 2354-1083.

8. ACADEMIC & PROFESSIONAL REFERENCES

1. **Assoc. Prof. Bui Van Hanh, Ph.D.**
    - Head of Department of Welding Engineering & Metals Technology, School of Mechanical Engineering, HUST
    - President, Vietnam Welding Society (VWS)
    - Email: hanh.buivan@hust.edu.vn | info@vws.org.vn
2. **Assoc. Prof. Pham Van Sang, Ph.D.**
    - Vice Dean, School of Mechanical Engineering, Hanoi University of Science and Technology (HUST)
    - Email: sang.phamvan@hust.edu.vn